CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

GenWise

…Wisdom through Genetics…

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A project report submitted in partial fulfillment of the requirements for the award of
Bachelors of Computer Applications of
CHRIST(Deemed to be University)
July 2026

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Introduction

1.1 Purpose

GenWise aims to make genetic counselling easy to understand, accessible, and supportive for families. The app helps people learn about their genetic risks by collecting family history, lifestyle details, and test results, then turning that information into clear insights. It connects users with certified counsellors through secure chat or video sessions, guiding them step-by-step from booking an appointment to reviewing results. It additionally offers practical tools to help families make informed choices about screening, prevention, and lifestyle changes. It includes a resource library with articles, FAQs, and visual guides, while AI adds smart recommendations and intuitive charts to explain inheritance patterns and risks.

1.2 Scope

The scope of the project include building a multi page website for Genwise that covers the following information:-

1. For Patients:
    - Home
    - About Us
    - Registration
    - Input Family Data
    - Risk Assessment
    - Schedule Counselling Sessions
    - Access Learning Materials
    - Memberships
    - Contact Us

2. For Counsellors:
    - Manage Appointments
    - Review Patient Data
    - Provide Treatment Recommendations

1.3 Stack Description

The project uses the following technologies for development:

- **Frontend:** HTML5, CSS3, JavaScript
- **Backend:** Node.js or Python Flask
- **Database:** MySQL or MongoDB
- **Security:** HTTPS, encryption, and role-based authentication

Overall Description

2.1 Requirement

The Genwise Web Application is required to provide a unified digital platform that supports both public visitors and registered patients. It creates a clear, patient-centered, and informative interface. The platform enables patients to explore genetic risk information, access educational resources, book counselling sessions, and securely communicate with counsellors. It also provides administrators and counsellors with tools to manage patient enquiries, update schedules, publish announcements, maintain resource libraries, and oversee content without requiring technical expertise. It addresses the growing need for accessible genetic counselling by integrating technology and healthcare. It helps users interpret genetic data, understand hereditary risks, and make informed health decisions.

2.2 Module Description

The GenWise Web Application is divided into several modules:

- **Home Page**: Overview, highlights, testimonials.
- **About Us:** Mission, vision, counsellor background.
- **Services:** Risk assessment, genetic testing guidance, results interpretation, psychosocial support.

- **Patient Journey**: Step-by-step counselling process.
- **Resources:** FAQs, guides, brochures, videos.
- **Appointments**: Online booking form.
- **Contact Us**: Inquiry form, contact details, clinic location.
- Dashboard: User data and personalized features.

2.3 Sub-Modules Description

- **Family History Input** – Collects hereditary data.
- **AI Risk Engine** – Calculates personalized risk scores.
- **Session Scheduler** – Books and tracks appointments.
- **Resource Library** – Displays curated educational content.
- **Admin Panel –** Admins can update schedules, manage content, and monitor enquiries.

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Functional Requirements

3.1. Software Requirements

- Server capable of hosting dynamic websites.
- Frontend: HTML, CSS, JavaScript.
- Backend: PHP.
- Database: MySQL for storing user data, schedules, and appointments.
- Compatibility with modern browsers (Chrome, Edge, Firefox, Safari

3.2. Hardware Requirements

- **Server:** Minimum 4 GB RAM, 2 CPU cores, 50 GB storage.
- **Development Machine:** 8 GB RAM, dual-core CPU, 20 GB storage.
- **Hosting:** Stable internet connection required.

3.3. User Requirements

- Users should be able to easily navigate across pages.
- Visitors can submit enquiries and requests using the contact form.
- Patients can book appointments and access personalized dashboards.
- Admins can monitor enquiries and update content.

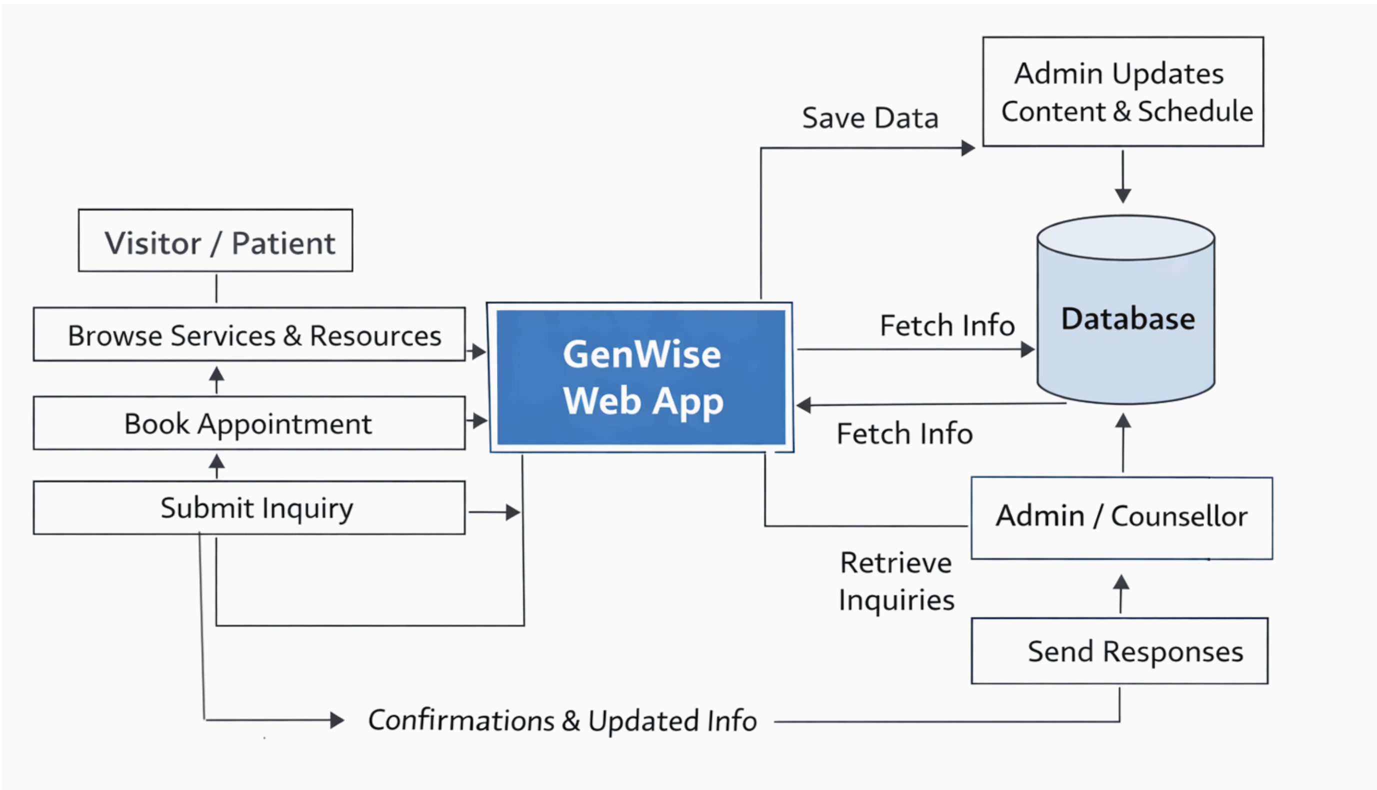
3.4 Functional Features

- User authentication and secure login
- Dynamic risk visualization charts
- Counsellor dashboard for patient management
- AI-based report summarization
- Notification and reminder system

Table Description (Database Schema)

Module	Sub-Module	Description	User Type
Home Page	—	Overview, highlights, testimonials, announcements	Visitor
About Us	—	Mission, vision, counsellor background	Visitor
Services	—	Details of counselling services	Visitor, Patient
Patient Journey	—	Step-by-step counselling process	Visitor, Patient
Resources	—	FAQs, guides, brochures, videos	Visitor, Patient
Appointments	Booking Form	Online appointment scheduling	Patient
Contact Us	Inquiry Form	Visitors can send enquiries	Visitor
Dashboard	Login/Signup	User account creation and secure login	Patient
Dashboard	User Profile	Manage personal info and preferences	Patient
Admin Panel	Content Management	Manage services, schedules, announcements	Visitor

Data Flow Diagram



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